SJ INNOVATION · ASSESSMENT 4 · ONE AI FEATURE, YOUR CHOICE

QuickBite — add one real AI feature

Food Ordering & Delivery · free tools only · one week · you pick the option

Team member	Sanket Rotangar
Mentor	Amol — Tech Manager
Issued	Thursday, 29 October 2026
Due	**Friday, 6 November 2026** (1 week) — submit on **Dev Control Tower**

Assessment 4 in one line: you've built the full app (A1–A3). Now add **one** genuinely useful **AI feature** to it, done properly and measured. **Choose one option below** — the choice is yours; pick the one you're most curious about. Everything here runs on **free tools only** (no paid APIs) and is scoped to **one week** using your existing seed data.

Choose one feature to build

Option 1 — Live predicted ETA + busy-kitchen signal (forecasting)

The feature you build: Replace the fixed delivery time with a predicted ETA and show a "kitchen busy" indicator.

What the output should look like: A customer sees a realistic, learned ETA (not a constant), and the app flags busy periods / overloaded branches. Success = the predicted ETA is close to the actual delivery time (MAE/RMSE reported).

AI concepts you'll explore: Forecasting / time series, regression evaluation (MAE/RMSE, backtesting), drift monitoring.

Build checklist:

- Build features from order history (time of day, day of week, branch load, distance).
- Train ETA/prep-time regression + a demand model (sklearn/XGBoost; Prophet).
- Report MAE/RMSE with backtesting; check leakage.
- Show the live ETA + busy indicator; monitor drift.

Free: sklearn / XGBoost / Prophet.

Option 2 — Find-dishes-by-photo + auto-tags (vision)

The feature you build: Let a customer search by a food photo and auto-tag menu items (cuisine / spicy / veg).

What the output should look like: A customer searches with an image and gets similar dishes; new menu items get tags suggested for the owner to accept. Success = the right dishes surface for a photo and the tags are accurate.

AI concepts you'll explore: Multimodal / Vision (CLIP image embeddings, image tagging), semantic search.

Build checklist:

- Embed dish images (CLIP / llava via Ollama) into pgvector.
- Build 'find dishes like this' + auto-tags.
- Owner review/correct screen.
- Measure search hit-rate + tag accuracy.

Free: CLIP / llava (Ollama), pgvector.

Option 3 — On-device AI with smart fallback (fine-tune + routing)

The feature you build: Run a small, fast model locally for your recommendations/classification and only call the cloud when it is unsure.

What the output should look like: The app answers with a fine-tuned local model first and escalates to Gemini only on low confidence; you show it is as accurate but cheaper and faster. Success = the local model matches or beats your A2 classifier (P/R/F1) with measured cost/latency saved.

AI concepts you'll explore: Fine-tuning (LoRA/QLoRA), distillation, local serving (Ollama), model routing / cascading.

Build checklist:

- Build a labelled set from your data.
- LoRA-fine-tune a small model on free Colab; export to Ollama.
- Compare local vs Gemini vs your A2 model (P/R/F1).
- Add routing: local first, escalate on low confidence; measure cost/latency.

Free: Colab (free T4 GPU), Ollama. (Good fit — you already run Ollama.)

How you will be scored (Assessment 4)

Parameter	Max
The feature works end-to-end, inside the app (not just a notebook/script)	25
Correct use of the AI concept/technique behind it	20
Evaluation & evidence (the right metric for this feature, with numbers)	20
Free-tools only, runs reliably (Gemini free / Ollama / open-source; no paid APIs)	10
UI/UX — how the feature is surfaced to the user	10
Tests, docs & Dev Control Tower submission	10
Short write-up: what the concept is, the trade-offs, and its limits	5
Total	**100**
Bonus — a fine-tuned / local model used, or you ship a second option	+10

What to submit on Dev Control Tower

1. Live URL + GitHub (tag an `assessment-4` release) and which option you chose.
2. A short demo (video or live) showing the feature working in the app.
3. Your evaluation numbers (the metric named in your option) + a one-paragraph write-up of the concept and its limits.
4. Updated test credentials.

> **Golden rule:** one feature, done properly and measured, beats a half-built idea. Keep it inside a week — free tools, your existing seed data.